AP PRECALCULUS

Unit 1 • Worksheet 1.2B • Answer key

Worksheet 1.2B • Rates of Change

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The table gives values of g. What is the average rate of change of g on $[2, 8]$?

x	2	4	6	8
$g(x)$	-3	1	6	9

(A) 12

(B) 2

(C) $\frac{1}{2}$

(D) 6

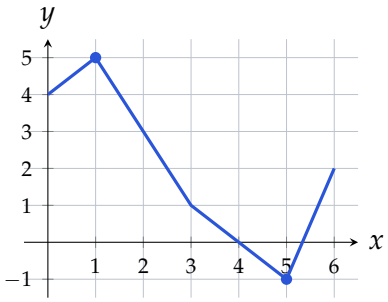
Solution

Answer: (B). $\frac{9-(-3)}{8-2} = \frac{12}{6} = 2$.

Question 2

NO CALCULATOR

The graph of h consists of line segments. What is the average rate of change of h on $[1, 5]$?



(A) $-\dfrac{3}{2}$
(B) -6
(C) $\dfrac{3}{2}$
(D) -1

Solution

Answer: (A). $h(1) = 5$, $h(5) = -1$: $\frac{-1-5}{4} = -\frac{3}{2}$.

Question 3

NO CALCULATOR

Using the same graph of h, on which interval is the average rate of change of h equal to -1?

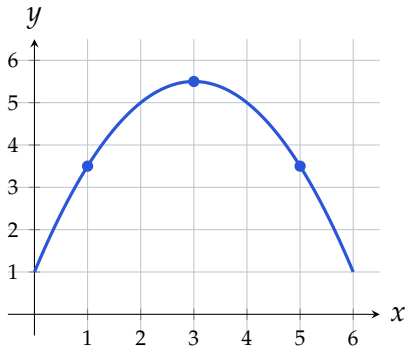
(A) $[0, 3]$
(B) $[3, 6]$
(C) $[1, 3]$
(D) $[0, 6]$

Solution

Answer: (A). $h(0) = 4$, $h(3) = 1$: $\frac{1-4}{3} = -1$. The others: $[3, 6]$: $\frac{1}{3}$; $[1, 3]$: -2; $[0, 6]$: $-\frac{1}{3}$.

Question 4 (Free response) NO CALCULATOR

The graph of f is shown.



(a) Find the average rate of change of f on $[1,3]$ and on $[3,5]$.
(b) Draw the secant line for $[1,5]$. What is its slope?
(c) Is the rate of change of f increasing or decreasing on $[0,6]$? What does this say about concavity?

Solution

(a) $f(1) = 3.5$, $f(3) = 5.5$, $f(5) = 3.5$. On $[1,3]$: $\frac{2}{2} = 1$. On $[3,5]$: $\frac{-2}{2} = -1$.
(b) The secant through $(1, 3.5)$ and $(5, 3.5)$ is horizontal: slope 0.
(c) Decreasing (the slopes go from positive to negative): the graph is concave down.